

Math Thermo
class 03
01/27/2025

Heat Flow

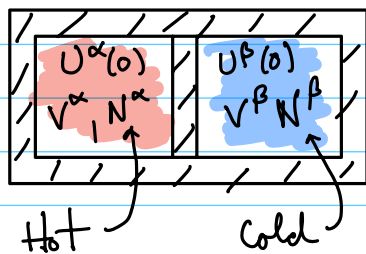
let us examine the approach to thermal equilibrium using a diathermal wall. Suppose the substance is unary, $r=1$. It must be that the variables

$$V^\alpha, V^\beta, N^\alpha, N^\beta$$

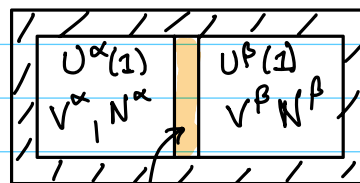
are fixed. But energy can be exchanged in the process.

Suppose the variable γ parameterizes the process

$$\gamma = 0$$



$$\gamma = 1$$



At $\gamma=0$,
we replace the
insulating wall
with a dia-
thermal wall

initial state

$$\begin{aligned} S(0) &= \tilde{S}^\alpha(U^\alpha(0), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(0), V^\beta, N^\beta) \\ &\leq \underbrace{\tilde{S}^\alpha(U^\alpha(1), V^\alpha, N^\alpha) + \tilde{S}^\beta(U^\beta(1), V^\beta, N^\beta)}_{\text{equilibrium state}} = S(1) \end{aligned}$$

We require that

$$U^\alpha(r) + U^\beta(r) = U_0$$

The entropy is at a max at equilibrium.

Suppose that, as indicated by the figure above,

$$T^\alpha(U^\alpha(0), V^\alpha, N^\alpha) > T^\beta(U^\beta(0), V^\beta, N^\beta)$$

we know that when equilibrium is attained

$$T^\alpha(U^\alpha(1), V^\alpha, N^\alpha) = T^\beta(U^\beta(1), V^\beta, N^\beta)$$

We will show that

$$U^\alpha(0) > U^\alpha(1) \quad (\text{energy is lost in } \alpha)$$

and

$$U^\beta(0) < U^\beta(1) \quad (\text{energy is gained in } \beta)$$

To see this compute

$$\begin{aligned} \frac{dS}{dr} &= \frac{\partial \tilde{S}^\alpha}{\partial U^\alpha} \frac{\partial U^\alpha}{\partial r} + \frac{\partial \tilde{S}^\beta}{\partial U^\beta} \frac{\partial U^\beta}{\partial r} \\ &= \frac{1}{T_s^\alpha} \frac{\partial U^\alpha}{\partial r} + \frac{1}{T_s^\beta} \frac{\partial U^\beta}{\partial r} \\ (3.1) \quad &= \left(\frac{1}{\tilde{T}_s^\alpha(r)} - \frac{1}{\tilde{T}_s^\beta(r)} \right) \frac{\partial U^\alpha}{\partial r} \end{aligned}$$

where

$$\tilde{T}_s^q(r) := T_s^q(U^q(r)), \quad q = \alpha, \beta.$$

We know that

$$S(0) \leq S(1).$$

We know that

$$\tilde{T}_s^\alpha(0) > \tilde{T}_s^\beta(0)$$

and

$$\tilde{T}_s^\alpha(1) = \tilde{T}_s^\beta(0).$$

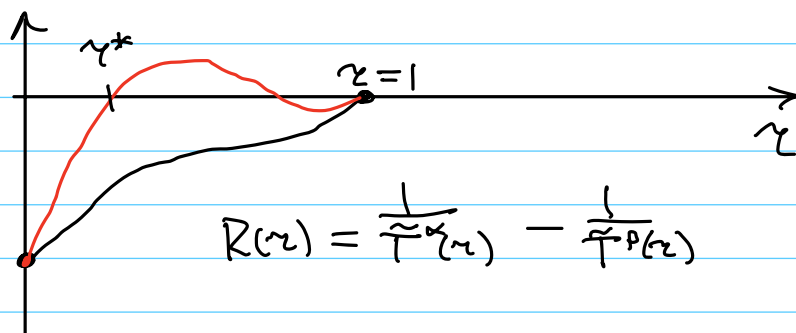
Set

$$R(r) := \frac{1}{\tilde{T}^\alpha(r)} - \frac{1}{\tilde{T}^\beta(r)} \quad 0 \leq r \leq 1$$

We know that

$$R(0) < 0 \quad \text{and} \quad R(1) = 0$$

Thus



The red curve is not possible; if it were equilibrium would be reached earlier, at $r=r^*$.

Thus

$$R(r) < 0 \quad \forall r \in [0, 1).$$

Integrating equation (3.1), we get

$$(3.2) \quad 0 \leq S(1) - S(0) = \int_0^1 R(r) \frac{dU^\alpha}{dr} dr$$

For $U^\alpha(r)$ we have two options

$$\text{Case (1)} \quad U^\alpha(0) > U^\alpha(1)$$

or

$$\text{Case (2)} \quad U^\alpha(0) \leq U^\alpha(1).$$

We are free to pick a parametrization however we want. Let us take a simple linear path

$$\frac{dU^\alpha}{dr} = \frac{U^\alpha(1) - U^\alpha(0)}{1} =: C^\alpha$$

Then

$$\text{Case (1)} \Rightarrow C^\alpha < 0$$

$$\text{Case (2)} \Rightarrow C^\alpha \geq 0$$

The integral in (3.2) is

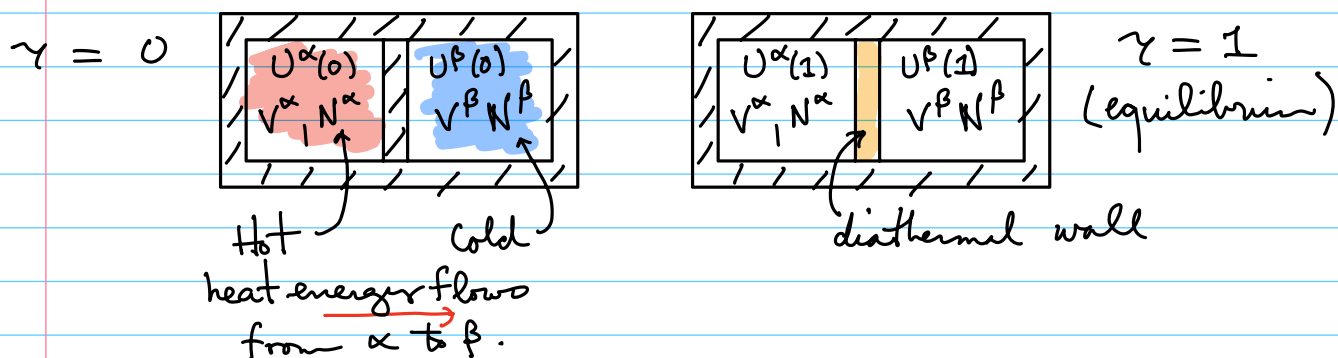
$$0 \leq S(1) - S(2) = C^\alpha \underbrace{\int_0^1 R(r) dr}_{< 0}.$$

Therefore, the only possible choice is Case (1).

Thus

$$U^\alpha(0) > U^\alpha(1) \quad (U^\beta(0) < U^\beta(1)).$$

This is consistent with our intuition about temperature. If $T^\alpha(0) > T^\beta(0)$ heat energy flow from subsystem α to subsystem β :



Heat Transfer Principle: Heat energy always flows from hotter to colder systems. This is equivalent to the second law of Thermodynamics.

The Euler Equation

Recall that internal energy and the entropy are homogeneous of degree one.

$$(3.3) \quad \text{where} \quad \tilde{U}(\lambda S, \lambda V, \lambda \vec{N}) = \lambda \tilde{U}(S, V, \vec{N}), \quad \lambda > 0,$$

$$\vec{N} = \begin{bmatrix} N_1 \\ N_2 \\ \vdots \\ N_r \end{bmatrix}.$$

Theorem (3.1): Let $\tilde{U}$ be the internal energy of an isolated system. Then

$$(3.4) \quad \tilde{U} = TS - PV + \mu_1 N_1 + \dots + \mu_r N_r$$

Proof: Since $\tilde{U}$ is homogeneous of degree one differentiating equation (3.1) with respect to λ

we obtain

$$\begin{aligned} T(\lambda S, \lambda V, \lambda \vec{N}) S - P(\lambda S, \lambda V, \lambda \vec{N}) V \\ + \sum_{j=1}^r \mu_j(\lambda S, \lambda V, \lambda \vec{N}) N_j = \tilde{U}(S, V, \vec{N}) \end{aligned}$$

Taking $\lambda=1$, we get

$$\begin{aligned}\tilde{U}(s, v, \vec{N}) &= T(s, v, \vec{N})S - P(s, v, \vec{N})V \\ &\quad + \sum_{j=1}^r \mu_j(s, v, \vec{N}) N_j,\end{aligned}$$

as desired. ///

Remark: Equation (3.4) is known as Euler's Equation.

Defn (3.2): A process path in state space Σ_S ,

$$\Sigma_S \subseteq [0, \infty) \times [0, \infty) \times [0, \infty) \times \cdots \times [0, \infty)$$

$\uparrow$
(S)

$\uparrow$
(V)

$\uparrow$
(N₁)

$\cdots$

$\uparrow$
(N_r)

is a continuous, piecewise differentiable function $\vec{\gamma}: [0, 1] \rightarrow \Sigma_S$, defined by

$$\vec{\gamma}(\gamma) = \begin{bmatrix} S(\gamma) \\ V(\gamma) \\ N_1(\gamma) \\ \vdots \\ N_r(\gamma) \end{bmatrix}.$$

A process path in Σ_V is defined similarly.

Thus, we have, using the chain rule,

$$(3.5) \quad \frac{d}{d\tau} \tilde{U}(\vec{\gamma}(\tau)) = T(\vec{\gamma}(\tau)) S'(\tau) - P(\vec{\gamma}(\tau)) V'(\tau) + \sum_{j=1}^r \mu_j(\vec{\gamma}(\tau)) N_j'(\tau)$$

for a valid process path in state space.

Theorem (3.3): Suppose that $\vec{\gamma}: [0,1] \rightarrow \Sigma_S$ is a process path in state space Σ_S . Then,

$$(3.6) \quad 0 = S(\tau) \frac{dT_0(\vec{\gamma}(\tau))}{d\tau} - V(\tau) \frac{dP_0(\vec{\gamma}(\tau))}{d\tau} + \sum_{j=1}^r N_j(\tau) \frac{d\mu_{0,j}(\vec{\gamma}(\tau))}{d\tau}.$$

This equation is called the Gibbs-Duhem relation.

Proof: Begin with the Euler equation and differentiate with respect to the process parameter τ :

$$(3.7) \quad \begin{aligned} \frac{d\tilde{U}}{d\tau}(\vec{\gamma}(\tau)) &= \frac{dT}{d\tau}(\vec{\gamma}(\tau)) S(\tau) + T(\vec{\gamma}(\tau)) S'(\tau) \\ &\quad - \frac{dP}{d\tau}(\vec{\gamma}(\tau)) V(\tau) - P(\vec{\gamma}(\tau)) V'(\tau) \\ &\quad + \sum_{j=1}^r \left\{ \frac{d\mu_j}{d\tau}(\vec{\gamma}(\tau)) N_j(\tau) + \mu_j(\vec{\gamma}(\tau)) N_j'(\tau) \right\} \end{aligned}$$

Subtracting (3.5) from (3.7), we get (3.6). |||

Example (3.4): Suppose that the fundamental relation for a material is given by

$$\tilde{S} = 4A U^{1/4} V^{1/2} N^{1/4} + BN, \quad \Sigma_U = [0, \infty).$$

where $A, B > 0$ are constants. This function must be homogeneous of degree one. let us check that. Suppose $\lambda > 0$. Then

$$\begin{aligned} \tilde{S}(\lambda U, \lambda V, \lambda N) &= 4A \lambda U^{1/4} \lambda V^{1/2} \lambda N^{1/4} + B \lambda N \\ &= \lambda \tilde{S}(U, V, N) \quad // \end{aligned}$$

Recall that

$$\begin{aligned} T &= \frac{1}{\frac{\partial \tilde{S}}{\partial U}} = \left(A U^{-3/4} V^{1/2} N^{1/4} \right)^{-1} \\ &= \frac{U^{3/4}}{A V^{1/2} N^{1/4}}, \end{aligned}$$

which is homogeneous degree zero.

$$\frac{P}{T} = \frac{\partial \tilde{S}}{\partial V}$$

So

$$P = T \frac{\partial \tilde{S}}{\partial V}$$

$$= \frac{U^{3/4}}{A V^{1/2} N^{1/4}} \left(\frac{2A U^{1/4} N^{1/4}}{V^{1/2}} \right)$$

$$= \frac{2U}{V}$$

Finally,

$$\mu = -T \frac{\partial \tilde{S}}{\partial N}$$

$$= \frac{-U^{3/4}}{A V^{1/2} N^{1/4}} \left(\frac{A U^{1/4} V^{1/2}}{N^{3/4}} + B \right)$$

$$= -\frac{U}{N} - \frac{B U^{3/4}}{A V^{1/2} N^{1/4}} \quad ///$$

Example (3.5): Suppose that the fundamental relation is

$$\tilde{U} = \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^4$$

Recall that

$$T = \frac{\partial \tilde{U}}{\partial S}$$

$$= 4 \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^3 \frac{1}{4 A V^{1/2} N^{1/4}}$$

$$= \frac{\tilde{U}^{3/4}}{A V^{1/2} N^{1/4}},$$

The same as above.

$$P = -\frac{\partial \tilde{U}}{\partial V}$$

$$= -4 \left(\frac{S - BN}{4 A V^{1/2} N^{1/4}} \right)^3 \left(\frac{S - BN}{4 A N^{1/4}} \right) \left(-\frac{1}{2} \right) \frac{1}{V^{3/2}}$$

$$= 2 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^4 \frac{1}{V}$$

$$= \frac{2\tilde{U}}{V}.$$

Finally,

$$\mu = \frac{\partial \tilde{U}}{\partial N}$$

$$= 4 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^3 \frac{1}{4AV^{1/2}} \frac{N^{1/4}(-B) - (S - BN)\frac{1}{4}N^{-3/4}}{N^{1/2}}$$

$$= 4 \left(\frac{S - BN}{4AV^{1/2}N^{1/4}} \right)^3 \frac{1}{4AV^{1/2}} \frac{N^{1/4}(-B) - (S - BN)\frac{1}{4}N^{-3/4}}{N^{1/2}}$$

$$= 4\tilde{U}^{3/4} \cdot \frac{N^{1/4}}{4AV^{1/2}N^{1/2}} \cdot \left(-B - \frac{S - BN}{4N} \right)$$

$$= \frac{-B\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} - \frac{\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} \cdot \frac{(S - BN)}{4N}$$

$$= \frac{-B\tilde{U}^{3/4}}{AV^{1/2}N^{1/4}} - \frac{\tilde{U}}{N} \cdot \text{///}$$

let us establish the Euler Equation with respect to entropy.

Theorem (3.5): Let $\tilde{S}$ be the internal energy of an isolated system. Then

$$(3.8) \quad \tilde{S} = \frac{1}{T_s} U + \frac{P_s}{T_s} V - \sum_{j=1}^r \frac{\mu_{s,j}}{T_s} N_j,$$

where

$$T_s = T_s(U, V, \vec{N}),$$

$$P_s = P_s(U, V, \vec{N}),$$

$$\mu_{s,j} = \mu_{s,j}(U, V, \vec{N}),$$

and

$$\frac{1}{T_s} = \frac{\partial \tilde{S}}{\partial U}, \quad \frac{P_s}{T_s} = \frac{\partial \tilde{S}}{\partial V}, \quad \frac{\mu_{s,j}}{T_s} = \frac{\partial \tilde{S}}{\partial N_j}.$$

Proof: We again use the fact that $\tilde{S}$ is homogeneous of degree one. For any $\lambda > 0$,

$$\tilde{S}(\lambda U, \lambda V, \lambda \vec{N}) = \lambda \tilde{S}(U, V, \vec{N})$$

Taking the derivative with respect to λ , we have

$$\begin{aligned} & \frac{1}{T_s(\lambda U, \lambda V, \lambda \vec{N})} U + \frac{P_s(\lambda U, \lambda V, \lambda \vec{N})}{T_s(\lambda U, \lambda V, \lambda \vec{N})} V \\ & + \sum_{j=1}^r \frac{\mu_{s,j}(\lambda U, \lambda V, \lambda \vec{N})}{T_s(\lambda U, \lambda V, \lambda \vec{N})} N_j = \tilde{S}(U, V, \vec{N}) \end{aligned}$$

setting $\lambda = 1$ gives the desired result. //

The Gibbs - Duhem equation is similarly derived.

Theorem (3.6): Suppose that $\vec{\gamma}: [0,1] \rightarrow \Sigma_0$ is a process path. Then

$$(3.9) \quad 0 = \frac{dT_s}{d\tau}(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) - \frac{dP_s}{d\tau}(\vec{\gamma}(\tau)) V(\tau) + \sum_{j=1}^r \frac{d\mu_{sj}}{d\tau}(\vec{\gamma}(\tau)) N_j(\tau).$$

This equation is called the Gibbs - Duhem relation in the entropy form.

Proof: Using (3.8), we have

$$T_s(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) = U(\tau) + P_s(\vec{\gamma}(\tau)) V(\tau) - \sum_{j=1}^r \mu_{sj}(\vec{\gamma}(\tau)) N_j(\tau).$$

Taking the τ -derivative of the last equation we have

$$(3.10) \quad \begin{aligned} & \frac{dT_s}{d\tau}(\vec{\gamma}(\tau)) \tilde{S}(\vec{\gamma}(\tau)) + T_s(\vec{\gamma}(\tau)) \frac{d\tilde{S}}{d\tau}(\vec{\gamma}(\tau)) \\ &= U'(\tau) + \frac{dP_s}{d\tau}(\vec{\gamma}(\tau)) V(\tau) + P_s(\vec{\gamma}(\tau)) V'(\tau) \\ & \quad - \sum_{j=1}^r \left\{ \frac{d\mu_{sj}}{d\tau}(\vec{\gamma}(\tau)) N_j(\tau) + \mu_{sj}(\vec{\gamma}(\tau)) N_j'(\tau) \right\} \end{aligned}$$

Taking the τ derivative of $\tilde{S}(\vec{\gamma}(\tau))$ we have

$$\frac{d\tilde{S}}{d\tau}(\vec{\gamma}(\tau)) = \frac{1}{T_s(\vec{\gamma}(\tau))} U'(\tau) + \frac{P_s(\vec{\gamma}(\tau)) V'(\tau)}{T_s(\vec{\gamma}(\tau))}$$

$$- \sum_{j=1}^r \frac{\mu_{s,j}(\vec{s}(r))}{T_s(\vec{s}(r))} N_j'(r) .$$

Equivalently ,

$$(3.11) \quad T_s(\vec{s}(r)) \frac{dS}{dr}(\vec{s}(r)) = U'(r) + P_s(\vec{s}(r)) V'(r) - \sum_{j=1}^r \mu_{s,j}(\vec{s}(r)) N_j'(r)$$

Subtracting (3.11) from (3.10) we get

$$\frac{dT_s}{dr}(\vec{s}(r)) \tilde{S}(\vec{s}(r)) = \frac{dP_s}{dr}(\vec{s}(r)) V(r) - \sum_{j=1}^r \frac{d\mu_{s,j}}{dr}(\vec{s}(r)) N_j(r) ,$$

which is the desired result. ///

Remark: Compare (3.6) and (3.9):

$$(3.6) \quad 0 = S(r) \frac{dT_0}{dr}(\vec{s}(r)) - V(r) \frac{dP_0}{dr}(\vec{s}(r)) + \sum_{j=1}^r N_j(r) \frac{d\mu_{0,j}}{dr}(\vec{s}(r)) ,$$

$$(3.9) \quad 0 = \frac{dT_s}{dr}(\vec{s}(r)) \tilde{S}(\vec{s}(r)) - \frac{dP_s}{dr}(\vec{s}(r)) V(r) + \sum_{j=1}^r \frac{d\mu_{s,j}}{dr}(\vec{s}(r)) N_j(r) .$$

These are essentially the same expressions!